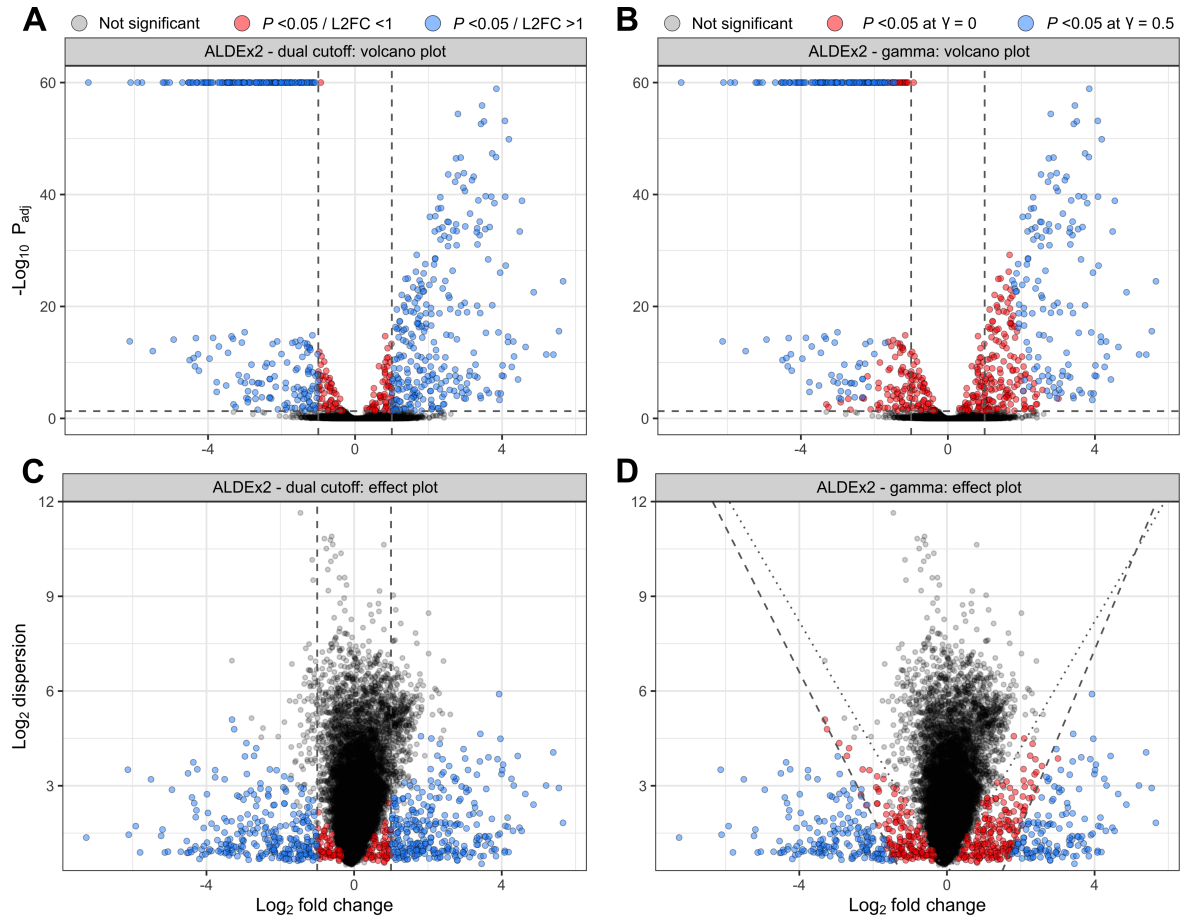




Supplementary Figure S4: ALDEx2 scale models provide better FDR control than a dual P -value/fold-change cutoff. A randomly selected analysis iteration of the PD1 immunotherapy dataset was used to create volcano plots (**A**, **B**) and effect plots (**C**, **D**) for ALDEx2 analyses. Each data point represents a feature from the scale-naïve analysis ($\gamma = 0.001$). Differentially expressed features are defined either by: a BH-adjusted P -value threshold of < 0.05 and a fold-change cutoff of ≥ 1 (**A**, **B**); or a BH-adjusted P -value threshold of < 0.05 when $\gamma = 0.5$ (**C**, **D**). Grey points indicate non-significant features. Coloured points indicate features which are significantly different between groups after applying different definitions of differential expression. Vertical dashed lines in **A-C** indicate a log_2 fold-change of 1; horizontal dashed lines indicate BH-adjusted P -value < 0.05 . Grey dotted and dashed lines in **D** indicate approximate isarithms between non-significant and significant features when $\gamma = 0.001$ and 0.5 , respectively.